For each scene i, which contains N_i QA pairs and 6 images, the score is computed as

$$Scenescore_i = \frac{1}{N_i} \sum_{j=1}^{N_i}$$

where $\overline{S}_i$ is the average image similarity over the 6 images in the scene i:

$$\overline{S}_i = \frac{1}{6} \sum_{k=1}^{6} S_{i,k}$$

And $S_{i,j}$ is defined as the **cosine similarity** between the original and adversarial image representations, normalized to $[0,1]$.

Here, $x_{i,j}$ denotes the **character length of the appended suffix** for the j-th question in scene i, which penalizes longer textual perturbations. To enforce the suffix-only constraint, a prefix-matching mechanism is applied: the submitted question must preserve the original question as its prefix. Any modification to the original prefix will result in a score of zero for that question.

The final score is obtained by averaging over all M scenes:

$$Final\ Score = \frac{}{M}$$